

Phase 1: Data Exploration and Preprocessing

A Comparative Study of RAG Techniques for Generating Image Captions

Group: G30
Assigned technique: Metadata-Filtered RAG
Dataset: MS COCO 2017
Module: Image Processing

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1 Introduction

Image captioning systems transform visual content into natural-language descriptions. Despite their usefulness, generated captions may omit relevant objects, confuse object categories, or mention content that is not present. Retrieval-Augmented Generation (RAG) addresses part of this problem by retrieving relevant examples and using their captions as supporting context.

Group G30 is assigned Metadata-Filtered RAG. The intended approach uses visual metadata, specifically object categories detected in an image, to constrain or re-rank the retrieval of reference images and captions. In the current implementation, YOLOv8n produces object labels and CLIP provides image and text representations. The present report documents Phase 1 only: dataset preparation, exploration, cleaning audit, metadata extraction, caption preprocessing, embedding/index artifacts, and integrity checks. Caption generation and comparative RAG experiments are not claimed as completed in this phase.

Numerical results are reported from the artifacts of the G30 repository (README, notebook, JSON statistics, manifests, split files, and the 16 figures). Where the repository does not contain enough evidence for a stronger claim, the limitation is stated explicitly.

2 Objectives and Scope

The objectives of Phase 1 were:

1. Explore the MS COCO object categories, object counts, scene complexity, and object co-occurrence.
2. Analyse the human-written captions, including length, vocabulary, part-of-speech distribution, and keyword-based diversity.
3. Audit image availability, image integrity, duplicates, and caption counts without silently changing the frozen dataset.
4. Establish disjoint knowledge-base, validation, and test splits.
5. Extract object metadata from images using YOLOv8n.
6. Preserve original captions while creating a lightly cleaned representation for indexing.
7. Prepare CLIP embeddings and FAISS retrieval indexes, as recorded in the supplied manifests.
8. Check dataset and artifact integrity and document anti-leakage controls.

The scope excludes model training, final caption generation, and the Phase 2/Phase 3 comparison of retrieval strategies.

3 Dataset and Experimental Splits

3.1 MS COCO 2017

The project uses MS COCO 2017 images, instance annotations, and human-written captions. The knowledge base and validation images are selected from train2017; test images are selected from val2017. This source separation supports evaluation without placing test images in the retrieval corpus.

3.2 Frozen split configuration

The repository contains the following ID lists. The files, rather than a newly generated random split, are treated as the source of truth. The split procedure uses Python's random shuffling

Table 3.1: Dataset partitions recorded in the repository.

Partition	Source	Images	Purpose
Knowledge base	train2017	10,000	Retrieval corpus
Validation	train2017	100	Configuration validation
Test	val2017	1,000	Final evaluation set

with seed 42. The first 10,000 shuffled training IDs form the knowledge base, the next 100 form validation, and the first 1,000 shuffled validation-source IDs form the test set. The repository reports zero pairwise split overlap.

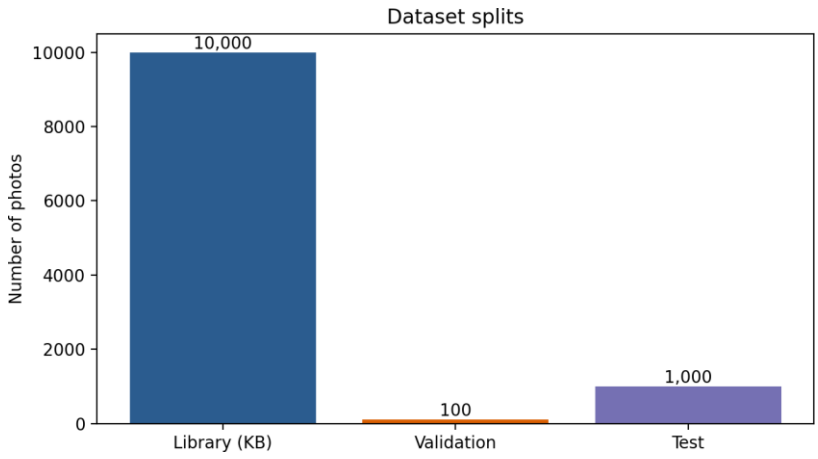


Figure 3.1: Number of images in the three frozen partitions.

3.3 Test-set size

The final test set contains **1,000 images** (test_ids.json). The repository also contains test_ids_100.json, an optional 100-image stratified subset selected from those 1,000 images (30 simple, 40 medium, and 30 complex images according to COCO object counts). This subset is provided only for optional quick experiments and is not the evaluation set: all results in this project are computed on the 1,000-image test set.

4 Data Audit and Cleaning

4.1 Audit methodology

The cleaning report covers the 11,100 images in the three partitions. It records issues rather than deleting images. This is important because removing an image after embeddings and indexes have been created would invalidate the frozen IDs and downstream artifacts. The audit checks:

- missing image files by mapping each ID to its expected COCO file;
- corrupted images using PIL verification and full decoding after EXIF orientation correction and RGB conversion;
- exact duplicates using MD5 file hashes;
- near-duplicates using perceptual hashes (pHash);
- duplicates across different splits;
- whether each image has the expected five captions.

4.2 Results

Table 4.1: Image and caption audit results from cleaning_report.json.

Check	Result
Missing images	0
Corrupted images	0
Exact duplicates	0
Near-duplicates (identical pHash)	0
Cross-split duplicate images	0
Images with fewer than five captions	0
Images with six captions	35
Total knowledge-base captions	50,031

The 35 images with six captions consist of 31 knowledge-base images, one validation image, and three test images. Consequently, the knowledge base contains 50,031 captions rather than exactly 50,000. No image or caption was removed by the audit. The additional captions are retained in the retrieval data. The repository also provides evaluation-reference files; the report does not assume that every image has exactly five source annotations.

The near-duplicate test is conservative: the reported zero refers to identical perceptual hashes. It should not be interpreted as proof that no visually similar images exist under a Hamming-distance threshold.

5 Exploration of Raw Data

5.1 Object categories

COCO annotations show that all 80 object categories occur in the 10,000-image knowledge base. The distribution is highly imbalanced. Person is the dominant category, with approximately 22,000 annotated instances; car, chair, bottle, and book follow at substantially lower frequencies. Approximately 53.98% of knowledge-base images contain at least one person.

5.2 Objects per image and scene complexity

The average number of annotated objects is 7.27 per image, while the median is 4. The distribution is right-skewed, with a long tail of images containing many objects. For analysis, images are grouped into four categories: none (0 objects), simple (1–3), medium (4–7), and complex (8 or more). About one third of the knowledge-base images are complex under this definition.

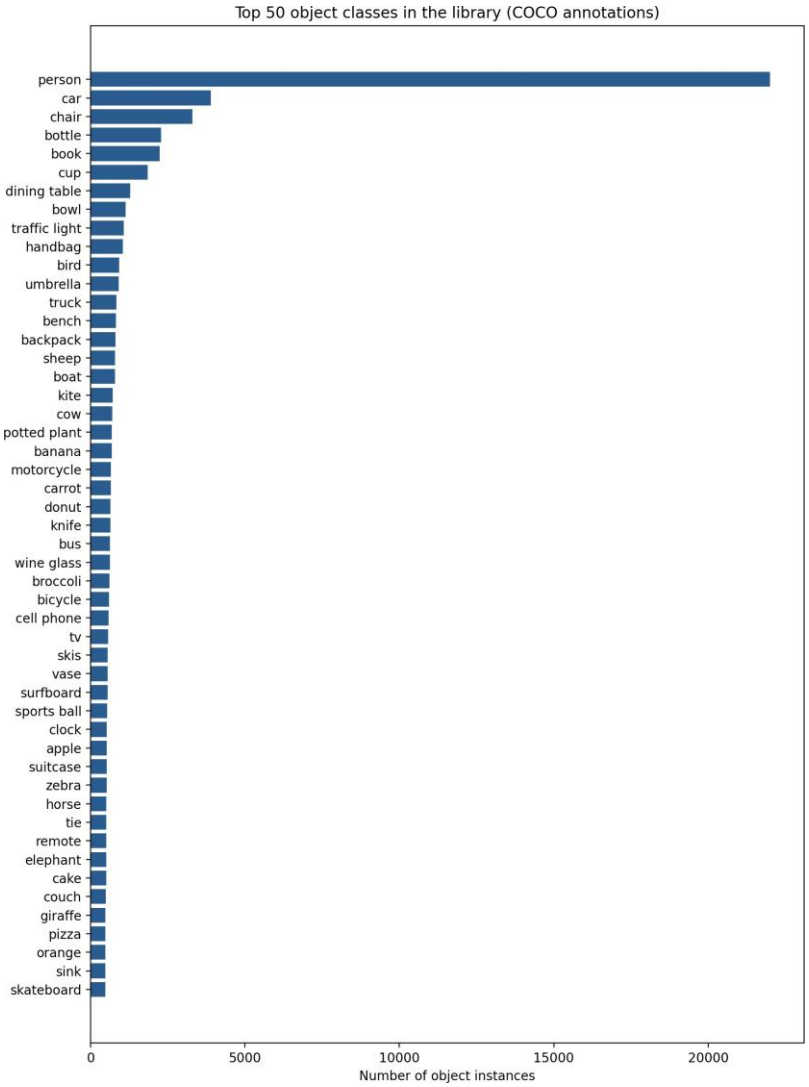


Figure 5.1: The 50 most frequent COCO object categories in the knowledge base.

Table 5.1: Scene complexity based on COCO annotations.

Scene category	Images	Share
None (0)	96	0.96%
Simple (1–3)	4,143	41.43%
Medium (4–7)	2,456	24.56%
Complex (8+)	3,305	33.05%

This is relevant because scenes containing many objects create a more demanding retrieval and captioning setting.

5.3 Object co-occurrence

The co-occurrence analysis considers the 20 most frequent categories. Person co-occurs with many categories, while pairs such as chair–dining table and cup–dining table reflect recurring indoor contexts. Co-occurrence indicates that object labels are not independent. In Phase 2,

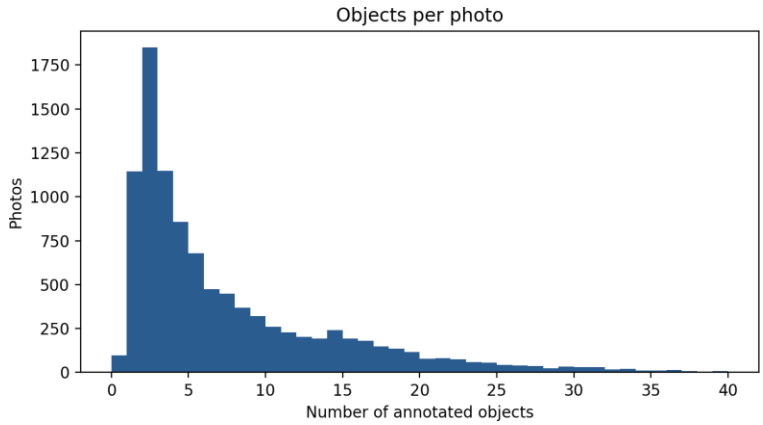


Figure 5.2: Distribution of annotated objects per knowledge-base image.

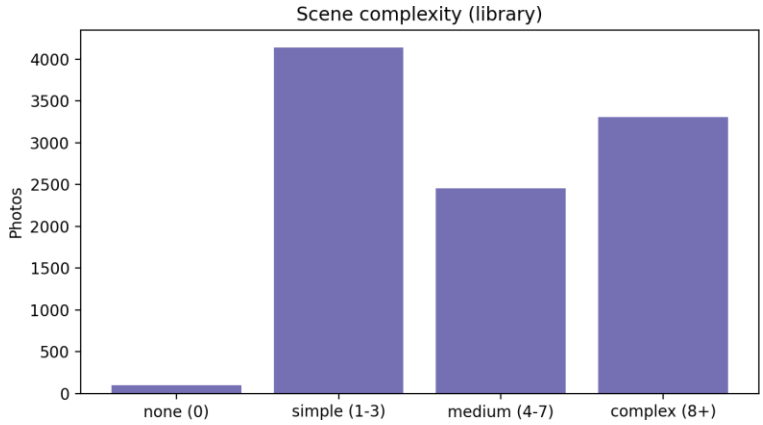


Figure 5.3: Scene-complexity distribution based on COCO annotations.

metadata matching may therefore benefit from considering combinations of labels rather than treating every detected category as an isolated signal.

5.4 Caption statistics

The knowledge base contains 50,031 raw captions. Their mean length is 10.45 words, the median is 10 words, and the standard deviation is approximately 2.40 words. The vocabulary contains 9,864 distinct lowercase tokens. The most frequent words include function words such as “a”, “on”, “of”, “the”, and “in”, followed by content words such as “man”, “sitting”, “standing”, and “people” (Figure 5.6).

5.5 Part-of-speech and keyword-based diversity

A seeded sample of 5,000 captions (52,069 tagged tokens) was processed with NLTK. Nouns represent 36.0% of tagged tokens, determiners 20.6%, prepositions 16.0%, verbs 11.5%, adjectives 8.1%, other tags 6.0%, numbers 1.3%, and adverbs 0.6% (Figure 5.7). The noun share supports the observation that captions are strongly object-centred.

Colour and context statistics are based on keyword lists in the captions, not on direct visual

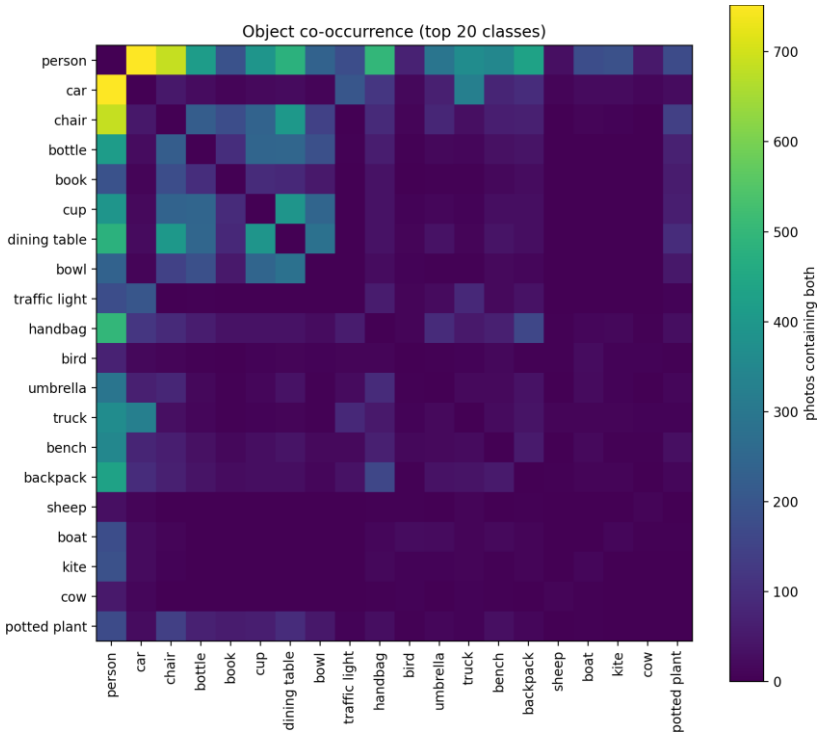


Figure 5.4: Co-occurrence matrix for the 20 most frequent COCO categories.

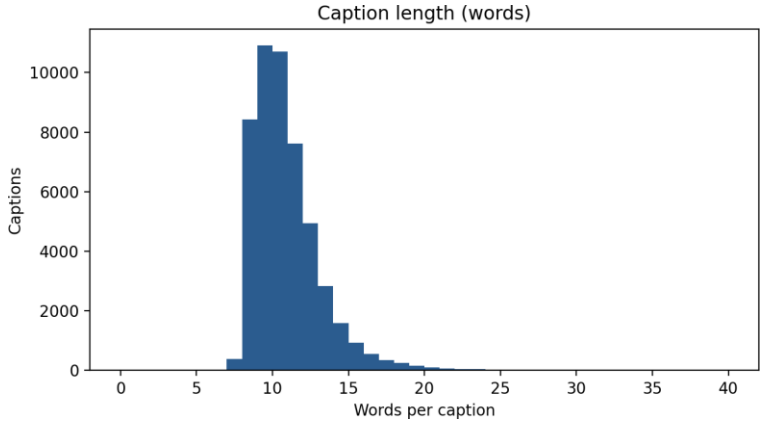


Figure 5.5: Length distribution of the original human-written captions.

classification (Figure 5.8). White is mentioned in 2,996 captions, black in 1,617, and red in 1,394. Outdoor keywords occur in 9,859 captions and indoor keywords in 6,838. Day and night terms are relatively rare (365 and 392 captions respectively). These values are indicative and depend on the selected keyword lists.

5.6 Spatial relationships

The repository's spatial analysis compares the eight largest annotated bounding boxes in each image. Relations are assigned geometrically: overlapping boxes, or left/right/above/below based on the dominant axis of centre displacement. Across 126,268 pairs, overlap accounts for 40,411 pairs (32.0%), left-of for 33,562 (26.6%), right-of for 33,144 (26.2%), below for 11,892

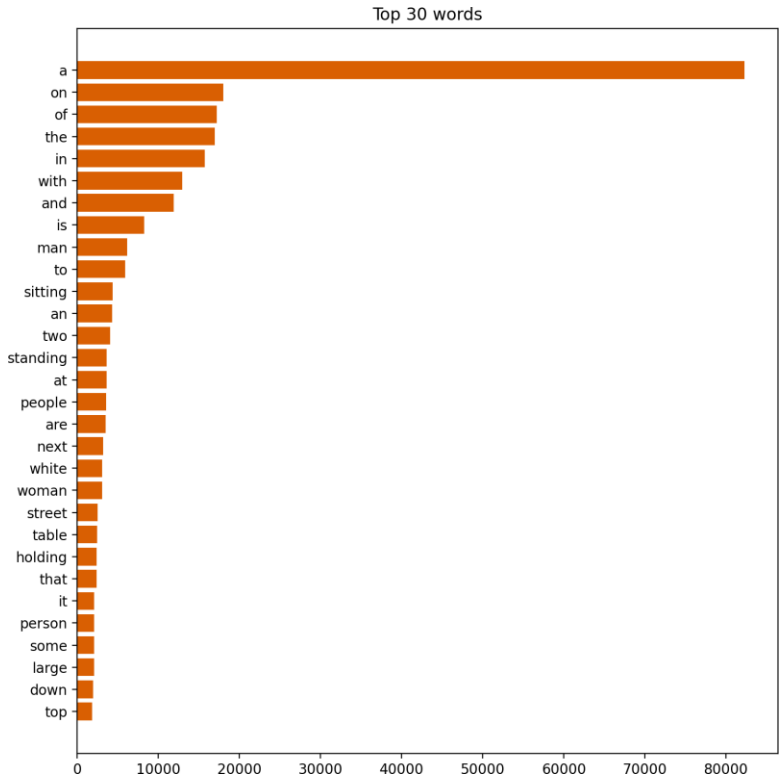


Figure 5.6: The 30 most frequent words in the knowledge-base captions.

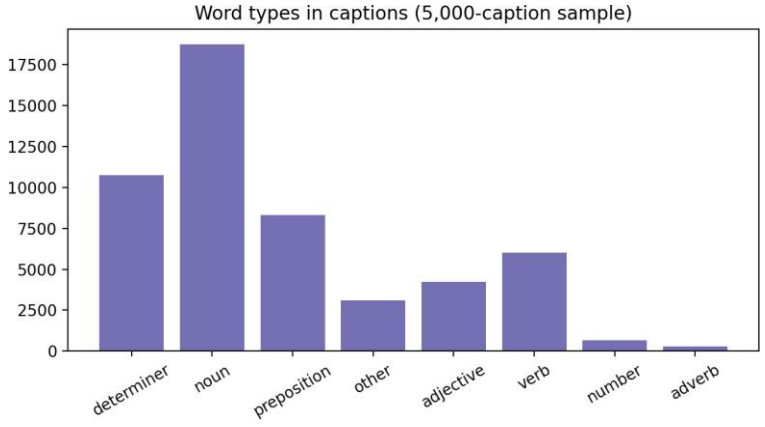


Figure 5.7: Part-of-speech distribution in a seeded sample of 5,000 captions.

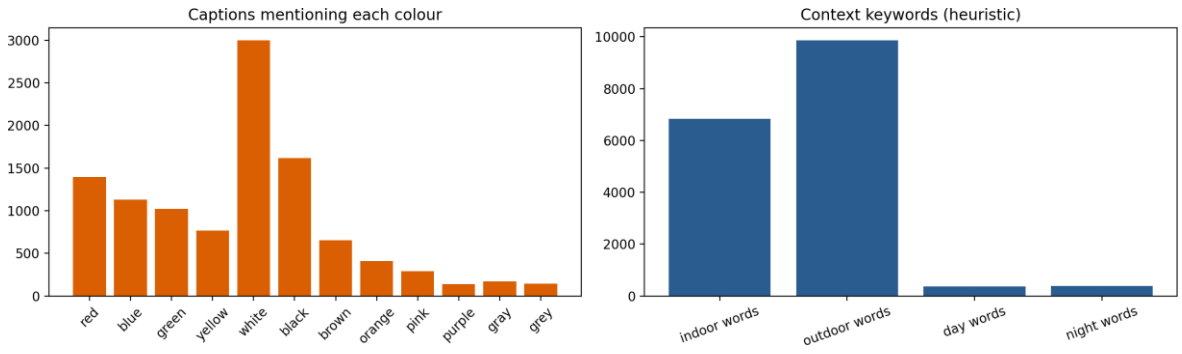


Figure 5.8: Colour and context keyword counts in the knowledge-base captions (keyword heuristic, not a scene classifier).

(9.4%), and above for 7,259 (5.7%) (Figure 5.9). These are geometric heuristics and should not be interpreted as semantic relations between objects.

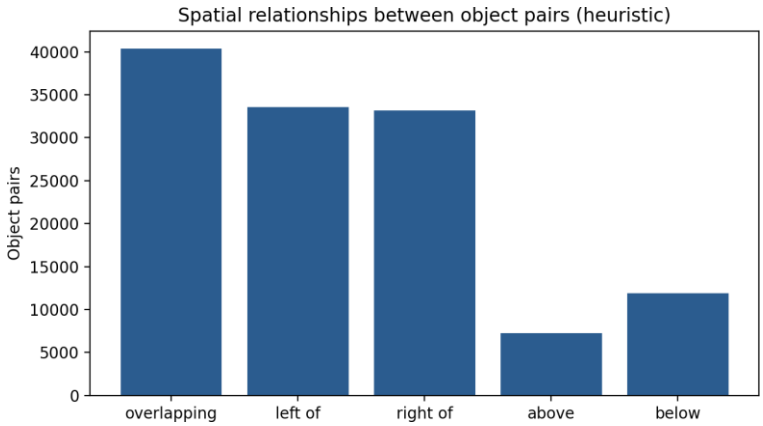


Figure 5.9: Geometric spatial relations between the eight largest annotated objects per image.

6 Image and Caption Preprocessing

6.1 Image preprocessing

The project uses PIL to open images, correct EXIF orientation, and convert them to RGB. The notebook's image pipeline prepares the input for the selected encoders. CLIP preprocessing follows the model processor; YOLO inference uses an input size of 640 pixels.

6.2 Caption preprocessing

Two caption representations are stored in kb_captions.jsonl:

- caption_raw: the original human-written caption, preserved for evaluation and traceability;
- caption_clean: the normalized version used for indexing and retrieval.

Cleaning is intentionally light: lowercase conversion, apostrophe removal, replacement of other punctuation with spaces, repeated-space collapse, and trimming. Digits are retained. The number of captions remains 50,031. Mean length changes from 10.45 to 10.46 words, and vocabulary size decreases from 9,864 to 9,762 (about 1.0%). No caption is discarded. The

Table 6.1: Raw and cleaned caption statistics.

Measure	Raw	Cleaned
Caption count	50,031	50,031
Mean length (words)	10.45	10.46
Vocabulary size	9,864	9,762

distinction between raw and cleaned text is essential: cleaning supports consistent retrieval, while evaluation must use the unchanged human references.

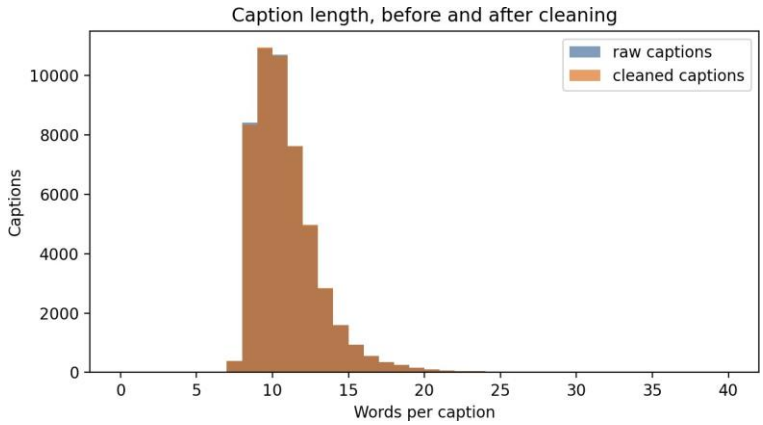


Figure 6.1: Caption length before and after normalization.

7 Visual Metadata Extraction

YOLOv8n (yolov8n.pt, Ultralytics 8.4.197) is used to extract object categories. The recorded configuration uses confidence threshold 0.25 and image size 640 for library, validation, and test images. The repository stores object names and the number of detections for each image. A separate optional notebook cell can store detection classes, confidence scores, and bounding boxes in JSONL format.

The metadata files cover 10,100 knowledge-base and validation images in yolo_metadata.json, and 1,000 test images in yolo_metadata_test.json. In the knowledge base, 177 images have no detected object. The detected object labels are included in the knowledge-base records and are the metadata fields intended for filtering or score boosting in the next phase. Attribute-level metadata such as colour or material and robust scene-level tags are not extracted in this phase.

7.1 Detector results

YOLO detects 79 categories in the library. The mean number of detections is 5.49 per image, the median is 4, and the maximum is 61. Person is detected in 5,215 images (52.15%). The next most common classes by number of images are car (918), dining table (917), chair (852), cup (660), and bottle (614).

7.2 COCO annotations compared with YOLO

Table 7.1: Human annotations and detector outputs on the 10,000-image knowledge base.

Measure	COCO annotations	YOLOv8n
Categories present	80	79
Mean objects/detections per image	7.27	5.49
Median	4	4
Images with no object	96	177
Images containing a person	53.98%	52.15%

YOLO produces fewer detections than the human annotations. The scene-complexity distribu-

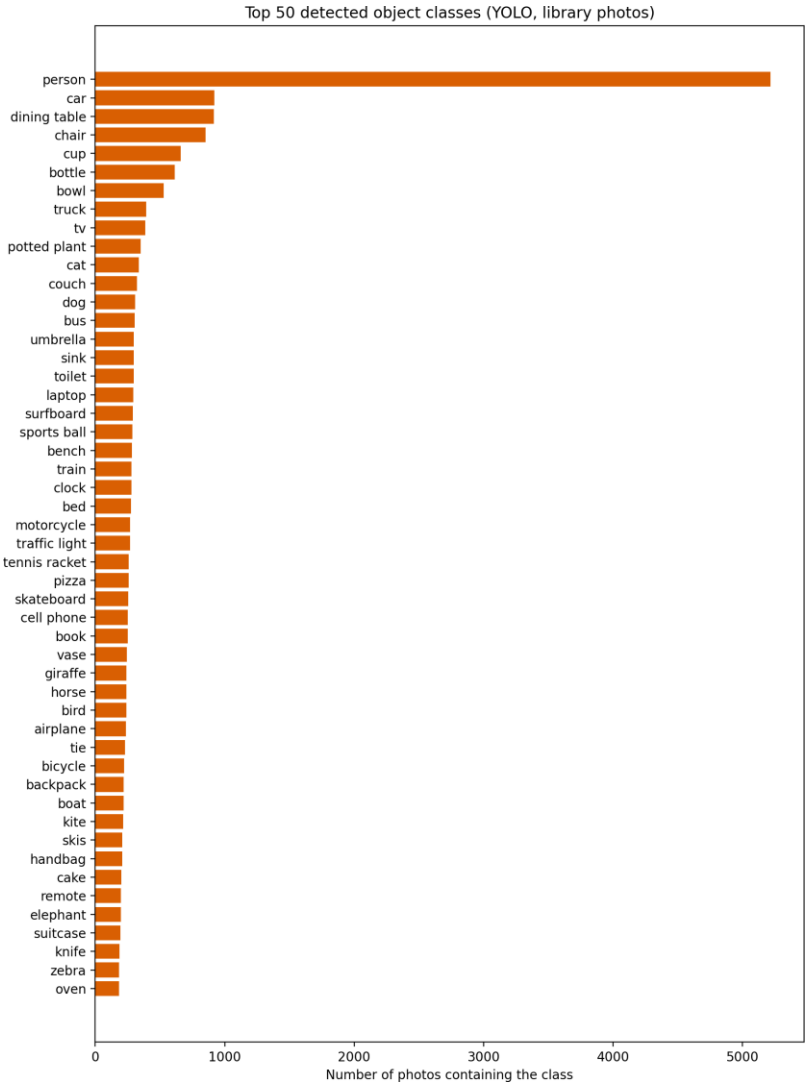


Figure 7.1: Most frequent YOLO-detected classes in the knowledge base.

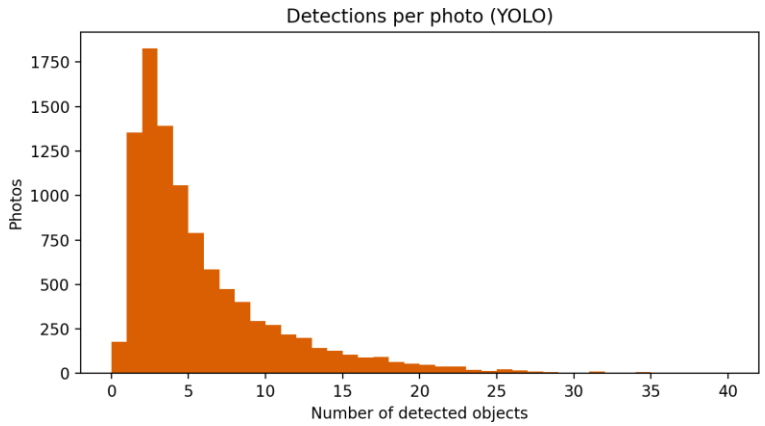


Figure 7.2: Number of YOLO detections per knowledge-base image.

tion consequently shifts: complex images decrease from 3,305 (33.05%) under COCO labels to 2,342 (23.42%) under YOLO detections. This difference illustrates that the detector output is an imperfect estimate of image content and that metadata-filtered retrieval may be affected by missed detections.

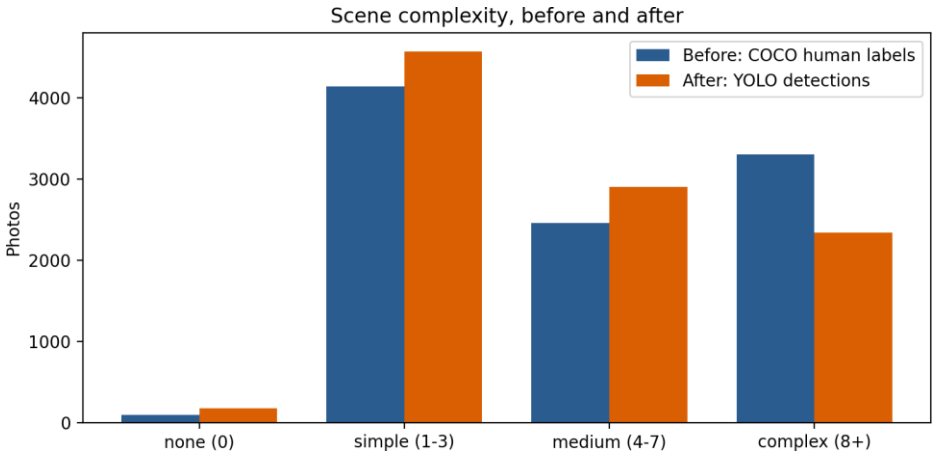


Figure 7.3: Scene complexity before and after object detection.

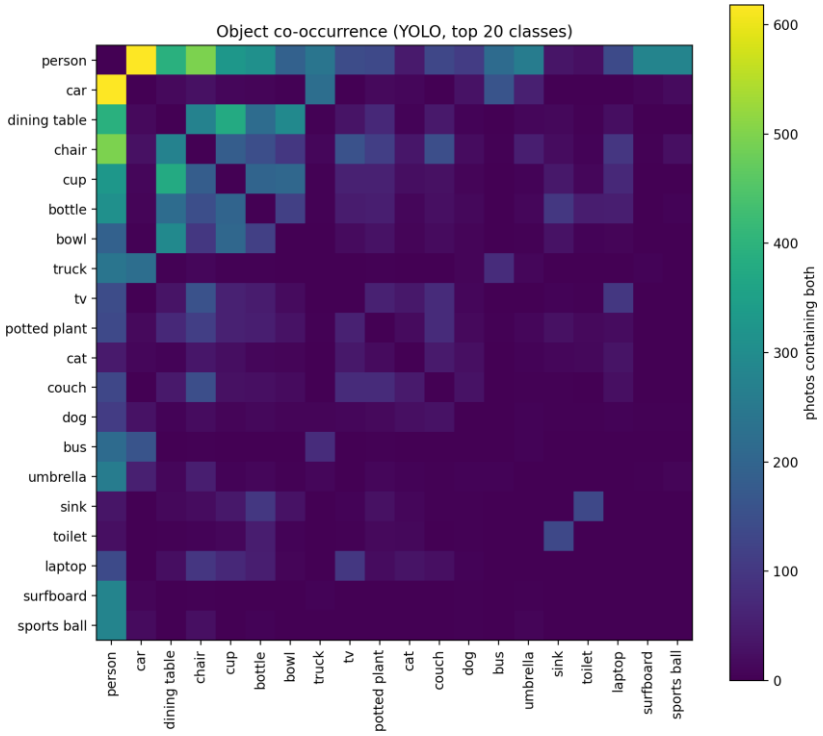


Figure 7.4: Co-occurrence matrix for the 20 most frequent YOLO classes.

7.3 Library–test comparison

The YOLO statistics indicate broadly similar scene distributions for the library and the 1,000-image test set. The mean detections per image are 5.49 and 5.45 respectively; both medians are 4. The proportions of simple, medium, and complex scenes are also close. This is a descriptive comparison, not proof that the two sets are identical in visual difficulty or distribution.

Table 7.2: YOLO-based scene statistics for the library and test set.

Measure	Library (10,000)	Test (1,000)
Mean detections	5.49	5.45
Median detections	4	4
Maximum detections	61	32
No detected object	1.77%	2.20%
Simple (1–3)	45.75%	45.40%
Medium (4–7)	29.06%	29.50%
Complex (8+)	23.42%	22.90%
Person present	52.15%	51.30%

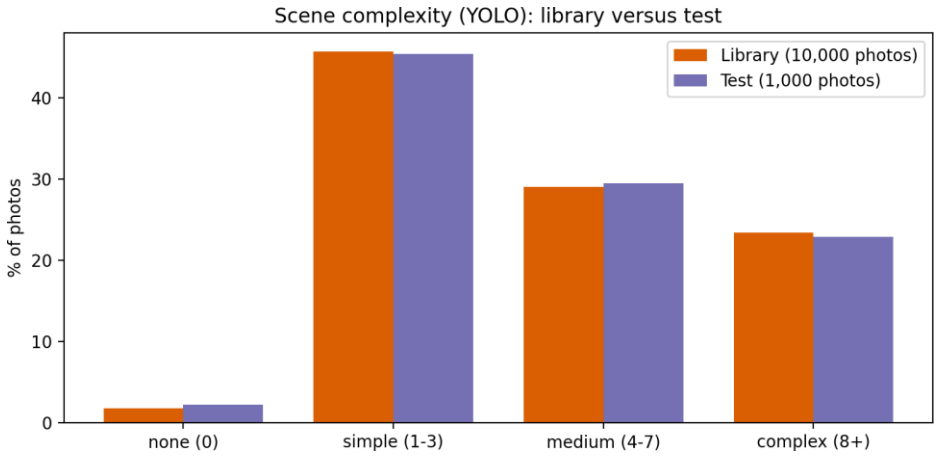


Figure 7.5: YOLO-based scene complexity in the library and test set.

8 Embeddings and Retrieval Indexes

The repository includes manifests describing the embedding and index artifacts. CLIP ViT-B/32 (openai/clip-vit-base-patch32) is used for both image and caption representations. The vectors have 512 dimensions, are stored as float32, and are L2-normalised. The manifests record 10,000 knowledge-base image vectors, 50,031 caption vectors, 100 validation image vectors, and 1,000 test image vectors.

The recorded generation environment uses PyTorch 2.11.0+cu130, Transformers 5.21.0, batch size 64, and CUDA. The manifest reports no failed images. Generation times recorded are 104.4 seconds for knowledge-base image vectors, 33.7 seconds for caption vectors, 1.5 seconds for validation image vectors, and 10.8 seconds for test image vectors.

FAISS indexes use IndexIDMap2(IndexFlatIP), an exact flat inner-product index with IDs. Because the vectors are L2-normalised, inner product is equivalent to cosine similarity. The index manifest reports 10,000 image vectors and 50,031 caption vectors; indexing time is 0.088 seconds. It also records self-retrieval@1 = 1.0 and image-to-caption hit@5 = 0.666. These are index/retrieval sanity indicators reported by the manifest; they are not caption-generation quality scores.

Reproducibility note. The submitted repository contains manifests and descriptions of the prepared artifacts, but it does not include the scripts that generated the embedding arrays and FAISS indexes. The report therefore documents the recorded configuration rather than claiming that embedding generation can be reproduced from the repository alone.

9 Anti-Leakage and Integrity Verification

9.1 Anti-leakage controls

The repository's caption audit reports zero pairwise image-ID overlap between knowledge base, validation, and test. It also reports zero overlap between knowledge-base captions and evaluation captions, zero duplicate caption IDs, and zero cross-split perceptual-hash matches. Validation and test images are not part of the knowledge-base search corpus. Test object metadata are produced by YOLO rather than by passing test COCO ground-truth labels to the retrieval system.

9.2 Artifact integrity audit

The notebook's A4 audit checks that required split, metadata, embedding, and index files exist and are non-empty; checks the expected split counts and pairwise ID disjointness; checks YOLO metadata coverage for knowledge-base and validation images; checks embedding shapes, float32 image embeddings, absence of NaN values, image-vector L2 normalisation, FAISS vector counts, and the caption-to-image mapping length. The notebook's documented success condition is the final message ALL CHECKS PASSED; the repository manifest also records leakage_audit_passed: true.

The A4 code does not itself implement every possible integrity assertion. In particular, its visible checks do not explicitly test duplicate IDs in every list, verify every caption-to-image entry against the knowledge-base record, or assert that no test ID appears in the caption-to-image map. These checks are planned before final evaluation. The caption audit JSON provides separate evidence for the reported caption and split-overlap checks.

10 Deliverables and Limitations

10.1 Phase 1 deliverables

The repository contains:

- frozen split ID files and the cleaning report;
- raw and cleaned knowledge-base captions and validation/test reference files;
- YOLO metadata, detector settings, and caption audit;
- CLIP and FAISS manifests, knowledge-base records, and caption-to-image mapping;
- JSON statistics, the 16 visualisation figures, and the Phase 1 notebook;
- README, requirements file, and team-member information.

The image files, embedding arrays, and FAISS binary indexes are not included in the GitHub-style repository archive because of their size; the README states that they are distributed separately through the team's shared archive.

10.2 Limitations

1. YOLOv8n is not ground truth. It detects fewer objects than the COCO annotations and may miss small or ambiguous objects.

2. Metadata currently consists primarily of object categories and detection counts. Attributes and robust scene/context labels are not available.
3. Colour, context, and spatial analyses use keyword or geometric heuristics and must be interpreted cautiously.
4. The near-duplicate audit only reports identical perceptual hashes.
5. The test set has 1,000 random images; the optional stratified 100-image subset exists but is not used for evaluation.
6. Embedding and index generation scripts are not included in the submitted repository, although their settings and results are recorded.
7. The A4 audit can be strengthened with explicit record-to-caption mapping and test-ID exclusion assertions.
8. The validation set has not yet been used for a documented confidence-threshold sweep in the supplied outputs.

11 Conclusion

Phase 1 has established a substantial data foundation for the G30 Metadata-Filtered RAG study. The repository records three disjoint partitions (10,000 knowledge-base images, 100 validation images, and 1,000 test images), a cleaning audit with no missing, corrupted, or duplicate images, 50,031 knowledge-base captions, YOLO object metadata, raw and cleaned caption representations, CLIP embeddings, and FAISS exact-search indexes.

The exploration indicates that the dataset is object-centred, that approximately one third of images are complex under the COCO annotation-based definition, and that object categories co-occur in recurring patterns. YOLO provides a practical source of metadata for the assigned method, but its lower detection count compared with human annotations is an important limitation. The current outputs are therefore a documented starting point for retrieval experiments on the 1,000-image test set. No claim of completed caption generation or final RAG performance is made in this Phase 1 report.